Andrew J. Miller

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EDUCATION

University of Colorado Boulder – B.S. Mechanical Engineering, Business Minor **Expected May 2028**
GPA: 3.9/4.0 | **Certifications:** CSWA; DeepLearning.AI: AI Agents in LangGraph, Prompt Engineering for Developers
Relevant Coursework: Experimental Methods, Circuits & Electronics, Computational Methods, Manufacturing Processes

TECHNICAL SKILLS

CAD & Design: SolidWorks (CSWA), 3D modeling, assemblies, sheet metal, drawings, tolerancing, GD&T, DFM
Fabrication: MIG welding, machining, soldering, laser cutting, FDM 3D printing (Bambu), assembly
Test & Analysis: Test design, sensor calibration, NI DAQ/LabVIEW, multimeter, oscilloscope, root cause analysis
Programming: MATLAB, Python, Arduino C++, embedded systems integration, quantitative data analysis
Artificial Intelligence: Prompt engineering, multi-model workflows, LLM-assisted automation pipelines and test plans

PROFESSIONAL EXPERIENCE

Teaching Assistant, Engineering Projects (GEEN 1400) – CU Boulder **Jan 2026 – Present**
- Guide student teams through the design, prototyping, and debugging of Arduino-based electromechanical projects that address real-world problems
- Plan and run lectures, labs, and workshops with faculty and fellow TAs, translating mechanical and electrical concepts for first-year engineers

Manufacturing Technician – Fiberlok Technologies, Fort Collins, CO **May – Aug 2023, 2024, 2025**
- Increased screens-department throughput 30% by overlapping emulsion coating with exposure and shortening dehaze and degrease cycles; method adopted into formal training
- Operated the screens department solo by the third summer, running a 6-step photopolymer process including per-job exposure settings and light-table inspection/QC
- Traced recurring print pinhole defects to residual emulsion, screen contamination, or hardened squeegee adhesive

Fabrication Intern – Hueber Industries, Fort Collins, CO **Jun – Aug 2024**
- Fabricated and installed functional and decorative metal structures, developing skills in welding, cutting, and forming while translating engineering drawings into builds held to 1/16 in tolerance
- Resolved a 250 lb railing installation failure by tracing it to assumed, inconsistent stair spacing, then re-measuring on site and reworking the railing in the shop

PROJECTS

ARC-001 Ultrasonic vs. Time-of-Flight Sensor Characterization – Personal Project **2026**
- Measured a 4x angular resolution advantage for VL53L1X time-of-flight over HC-SR04 ultrasonic across a six-target scene, tracing large ultrasonic errors to beam geometry, target orientation, and porosity
- Designed and FDM-printed a dual-sensor bracket in SolidWorks that stacked both sensors on one servo pan platform, pairing every reading of the same scene at the same instant
- Calibrated the rig as an instrument before data collection, deriving per-direction servo offsets from 1.43° measured backlash and scale corrections from a 14-station range ladder
- Developed the Python capture pipeline with an LLM under direct review, verifying its output and correcting 12 firmware and pipeline defects, including a silent 38 mm sensor offset

Bus Capacity Tracking System – CU Boulder, GEEN 1400 **Spring 2025**
- Led design and manufacturing of an Arduino-driven IR system that tracked and displayed bus occupancy; won section best project award
- Modeled transmitter and display housings in SolidWorks Sheet Metal with detailed drawings, and prepared manufacturable layouts in CorelDRAW
- Reached 98% detection accuracy through structured validation testing and iterative electrical and mechanical redesign

ENGINEERING TEAMS & ACTIVITIES

Team Member, EV Powertrain & Chassis – Formula SAE CU Boulder Racing Team **Sep 2026 – Present**
- Lathed two aluminum bolt adapters from engineering drawings and mounted damper linear potentiometers on the IC car, feeding suspension displacement to the data logger
- Assigned design of the regenerative brake pedal; currently in preliminary design